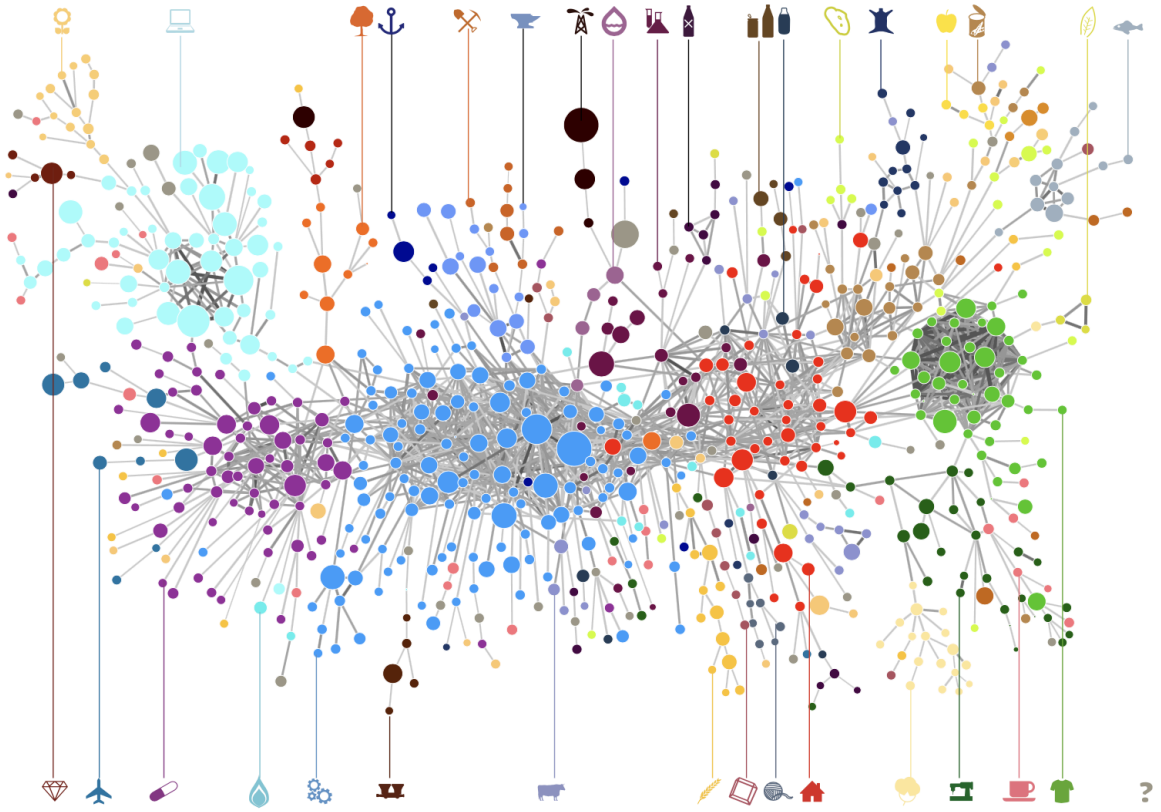


https://atlas.cid.harvard.edu/

Optimal Industry Development Strategy Using Economic and Product Complexity

Germany

—

The Economic Complexity Lab - Columbia University

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Optimal Industry Development Strategy for Germany

0. Executive Summary

In this report, we analyze the economic trajectory of Germany over the period from 2019 to 2024, examining its production capability and structural evolution through the lenses of Gross Domestic Product (GDP) and Economic Complexity Index (ECI). We find that Germany demonstrated a medium level of economic growth, with an average annual GDP growth rate of 3.0% during this period, placing it below its developed peers.

Germany's 2019–2024 trajectory reflected the strengths and constraints of a highly industrialized, export-oriented economy centered on machinery, transportation equipment, chemicals, and other complex manufacturing sectors.

On the complexity front, Germany's ECI increased over the analysis period, reflecting a standing below the peer average among the comparable developed economies examined. This ECI improvement only partially aligned with the GDP trend: while Germany's GDP rose sharply, ECI growth remained constrained.

Examining realized industry development between 2019 and 2024, we find that Germany regressed across the majority of its industry groups: 0 groups moved closer to global competitiveness thresholds while 18 retreated, a decrease from 16 improving groups at the start of the period. Based on our opportunity gain and sensitivity analysis, the industries offering the greatest complexity upside for Germany are Energy Drilling and Mining, Weapons, and Services and Utilities. Sensitivity analysis further identifies Transportation as the sector whose marginal expansion yields the largest positive shift in ECI.

Our Structural Optimality Index (SOI) analysis measures the degree to which Germany's realized investment trajectory aligns with the theoretically optimal complexity-based development path. Germany ranks in the Q4 quartile globally, with an adjusted R^2 of 0.0036, indicating a high level of structural alignment. In concrete terms, Germany has lightly channeled resources toward None — sectors where the complexity opportunity is more limited — while underinvesting in Chemicals, Wood Products, Metals, Machinery and Construction, High-complexity Products, Transportation, Weapons, Daily Instruments, Services and Utilities, which offer greater complexity upside. Given the positive correlation between SOI and long-run GDP growth established across our global sample, Germany's moderate structural alignment is consistent with its medium growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

1. Report Overview

In this report, we analyze the economic trajectory of Germany during the period from 2019 to 2024. This period was significant for Germany because it combined post-pandemic recovery, the 2022 energy shock, and growing pressure on Germany’s traditional export-manufacturing model, especially in transportation, machinery, chemicals, and energy-intensive industries. Specifically, we examine Germany's macroeconomic trajectory through the lens of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect a country's production capability and productive structure, respectively. We then analyze the industrial framework and structural changes of Germany based on quantitative measures including the Product Complexity Index (PCI), followed by a detailed analysis of how those changes aligned with optimal development strategies as measured by opportunity gains at the industry level. We conclude with macro-level results derived from regression models established on a global scale, which take into account countries' positions in the product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

Figure 1 illustrates the GDP time series for Germany and its peer economies (France, United Kingdom, and Italy) from 2019 to 2024. Germany demonstrated a medium average annual GDP growth rate of 3.0% over this period, placing it below peers among the compared economies. For reference, the average annual growth rates of the peer economies were: France at 2.7%, United Kingdom at 4.5%, and Italy at 3.0%.

Figure 1 shows Germany remaining the largest economy among the peer group throughout 2019–2024, with France and the United Kingdom forming a second tier and Italy and the Netherlands remaining lower.

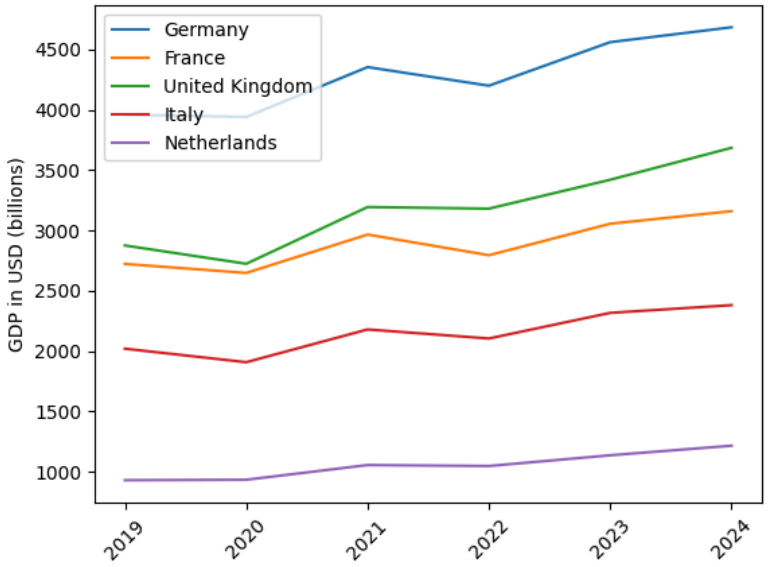


Figure 1. GDP Comparison Among Peer Countries (2019-2024)

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), introduced by Hausmann and Hidalgo (2009) to quantify the knowledge embedded in a country's productive structure, is illustrated in Figure 2.

Germany's ECI increased from -2.834 in 2019 to 1.309 in 2024, a change of +4.144 — a significant shift in productive complexity. Relative to its developed peers, Germany's ECI trajectory was below the peer average. For context, ECI changes over the same period were: France at +3.682, United Kingdom at +5.268, and Italy at +4.096.

The ECI movement appears to reflect broad repositioning across several complex industry groups rather than a single-sector effect. This interpretation is consistent with the later SDV and sensitivity visuals, where Transportation, Machinery and Construction, Metals,

Chemicals, Weapons, Services and Utilities, and Energy Drilling and Mining all appear as important sectors for Germany's complexity trajectory.

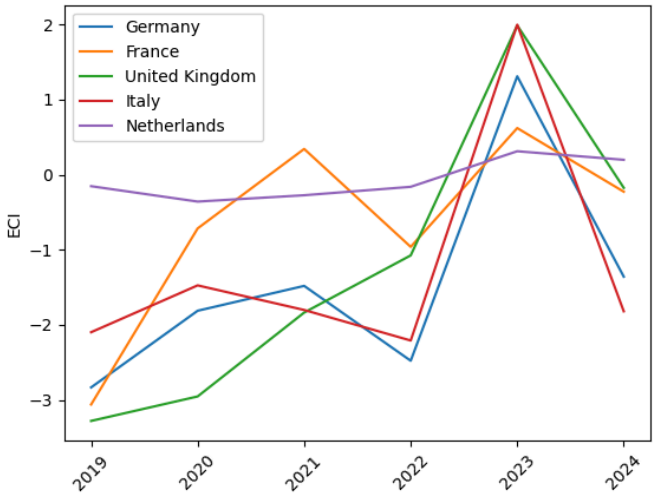


Figure 2. ECI Comparison Among Peer Countries (2019-2024)

Figure 3 provides a combined view of GDP and ECI for Germany across the full analysis window. The two indicators were partially consistent during this period. Figure 3 suggests that GDP and ECI moved in the same broad direction but not at the same pace: GDP followed a smoother upward path, while ECI showed sharper swings, indicating that Germany's structural complexity changed more unevenly than its aggregate output.

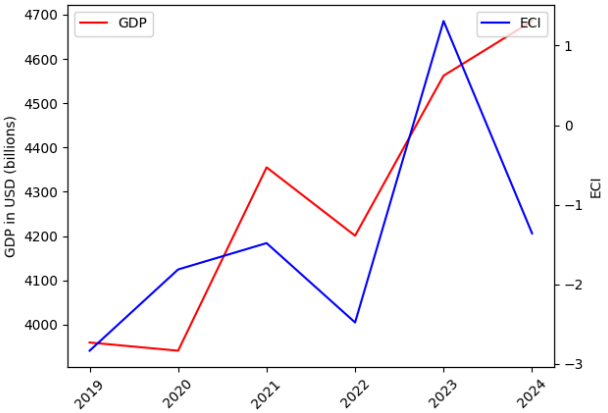


Figure 3. GDP Growth and ECI for Germany (2019-2024)

3. Production Analysis

3.1 Product Categorization

Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to

have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99 categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

Category	Description
Animals and By-products	All kinds of meat, livestock and related by-products
Vegetables, Fruits and Plants	All kinds of plants, edible vegetables and fruits
Food and Preparations	Edible preparations, raw cooking materials and appliances
Clothing and Accessories	All kinds of fabrics, filament, clothing and accessories
Textiles	All kinds of textile articles and related products
Wood Products	Book, paper and other elementary products related to wood
Metals	Mining of all sorts of raw Metals
Refined and Processed Stones	Refined or processed stone products such as pearl and ceramics
Energy Drilling and Mining	Mining industry for energy, minerals, stone materials, etc.
Daily Instruments	Basic daily instruments such as sport requisites, timing and measuring tools
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Chemicals	Raw chemical materials and by-products
Machinery and Construction	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapons	Explosives, arms and ammunition
Services and Utilities	Financial, business, legal, communication, travel Services and Utilities, etc.
High-complexity Products	Agglomeration of advanced products with highest PCIs

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economics, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries respectively.

	High-complexity Products	**Food and Preparations**	**Animals and By-products**
Adj. R-squared	0.075	0.006	0.002
F-statistic	9.556	1.638	1.175
No. Observations	107	107	107

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.149	0.003	-0.012
F-statistic	4.510	1.059	0.7542
No. Observations	21	21	21

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to capture Germany's realized industry production strategy between 2019 and 2024 on a complexity level. In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product, not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop the corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is of developing products within that group. Defining this net distance change as the Realized Industry Development (RID), our export data allow us to calculate and compare the RID of Germany and the 18 product groups in 2024 and 2022 through the difference, where a positive RID indicates a shortening of the distance (i.e., advancement of the industry), and a negative RID indicates the reverse. It is notable that the number of industries with positive RID experienced a decrease from 16 to 0 from 2022 to 2024.

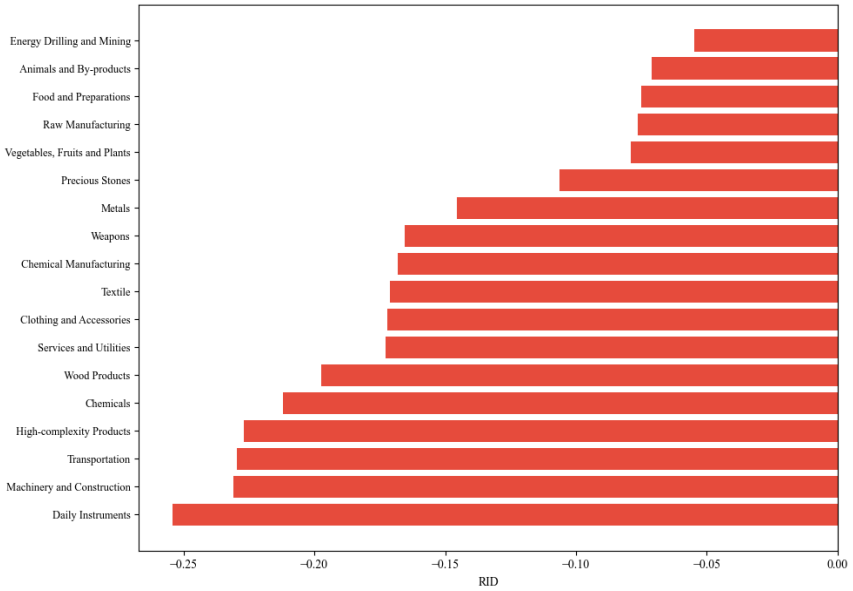


Table 3. Realized Industry Development (RID) of Germany

3.3 Regional and Global Comparison

Table 4a below summarizes how the production strategy adopted by Germany compares with selected leading economies in the world across different years, as measured in RID. Leading economies such as France, Greece, Portugal gave a considerable level of attention to more advanced product groups such as Energy Drilling and Mining in 2022.

Rank	France	Greece	Portugal
#1	Precious Stones	Chemicals	Precious Stones
#2	Wood Products	Weapons	Chemical Manufacturing
#3	Vegetables, Fruits and Plants	Chemical Manufacturing	Wood Products
#4	Food and Preparations	High-complexity Products	Energy Drilling and Mining
#5	Energy Drilling and Mining	Clothing and Accessories	Food and Preparations

Table 4a. RID Comparison across Leading Economies (2024)

Table 4b further examines Germany's development strategy against those of emerging economies on a global scale. Reasonably, emerging economies such as Poland, Czechia, and Turkey have devoted the majority of their efforts to intermediate industry groups, showing moderate shifts in strategies over the course of the analysis period.

	Poland	Czechia	Turkey	
	inery and Construction	Chemicals	Food and Preparations	High-complex
	h-complexity Products	Precious Stones	Energy Drilling and Mining	Vegetables, Fruit
	Daily Instruments	Wood Products	Clothing and Accessories	
	Chemicals	Metals	Animals and By-products	Machinery and
	Services and Utilities	Energy Drilling and Mining	Wood Products	Raw Ma

Table 4b. RID Comparison across Emerging Economies (2024)

4. Product Space Design

4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrement in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

Germany	France	United Kingdom	Italy	Netherlands
Transportation	Energy Drilling and Mining	Raw Manufacturing	Vegetables, Fruits and Plants	Precious Stones
Wood Products	Precious Stones	Machinery and Construction	Energy Drilling and Mining	Textile
Metals	Wood Products	High-complexity Products	Animals and By-products	Wood Products
High-complexity Products	Clothing and Accessories	Services and Utilities	Precious Stones	Services and Utilities
Daily Instruments	Weapons	Transportation	Food and Preparations	Clothing and Accessories
Services and Utilities	Machinery and Construction	Chemicals	Chemical Manufacturing	Animals and By-products
Machinery and Construction	Animals and By-products	Weapons	Raw Manufacturing	Vegetables, Fruits and Plants
Chemicals	Vegetables, Fruits and Plants	Clothing and Accessories	Textile	Food and Preparations
Weapons	Food and Preparations	Chemical Manufacturing	Clothing and Accessories	Energy Drilling and Mining
Clothing and Accessories	Chemical Manufacturing	Daily Instruments	Metals	Chemicals
Chemical Manufacturing	Textile	Textile	Machinery and Construction	Chemical Manufacturing
Textile	Metals	Wood Products	High-complexity Products	Raw Manufacturing
Raw Manufacturing	High-complexity Products	Metals	Daily Instruments	Weapons
Energy Drilling and Mining	Services and Utilities	Precious Stones	Wood Products	Daily Instruments
Precious Stones	Transportation	Energy Drilling and Mining	Services and Utilities	Machinery and Construction
Animals and By-products	Chemicals	Animals and By-products	Weapons	Metals
Food and Preparations	Daily Instruments	Food and Preparations	Transportation	Transportation
Vegetables, Fruits and Plants	Raw Manufacturing	Vegetables, Fruits and Plants	Chemicals	High-complexity Products

Table 5. Opportunity Gain Ranking on a Comparable Scale (2024)

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 summarizes how the SDV of Germany compares with a selected group of comparable and leading economies in the world, with the industry groups ranked in decreasing order of significance.

Dark blue indicates positive opportunity gains, light blue indicates zero opportunity gains, and white indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain and share common interests in Transportation and Wood Products industries. It can be concluded that Germany shares the characteristics of developing countries, with a balanced economic structure that leans towards intermediate-to-high-complexity industries.

France	United Kingdom	Italy	Netherlands
Energy Drilling and Mining	Raw Manufacturing	Vegetables, Fruits and Plants	Precious Stones
Precious Stones	Machinery and Construction	Energy Drilling and Mining	Textile
Wood Products	High-complexity Products	Animals and By-products	Wood Products
Clothing and Accessories	Services and Utilities	Precious Stones	Services and Utilities
Weapons	Transportation	Food and Preparations	Clothing and Accessories
Machinery and Construction	Chemicals	Chemical Manufacturing	Animals and By-products
Animals and By-products	Weapons	Raw Manufacturing	Vegetables, Fruits and Plants
Vegetables, Fruits and Plants	Clothing and Accessories	Textile	Food and Preparations
Food and Preparations	Chemical Manufacturing	Clothing and Accessories	Energy Drilling and Mining
Chemical Manufacturing	Daily Instruments	Metals	Chemicals
Textile	Textile	Machinery and Construction	Chemical Manufacturing
Metals	Wood Products	High-complexity Products	Raw Manufacturing
High-complexity Products	Metals	Daily Instruments	Weapons
Services and Utilities	Precious Stones	Wood Products	Daily Instruments
Transportation	Energy Drilling and Mining	Services and Utilities	Machinery and Construction
Chemicals	Animals and By-products	Weapons	Metals
Daily Instruments	Food and Preparations	Transportation	Transportation
Raw Manufacturing	Vegetables, Fruits and Plants	Chemicals	High-complexity Products

Table 6. SDV of Other Sample Countries (2024)

4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\underset{x \in \mathbf{R}^{|P|}}{\text{Max}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

$$\text{Subject to}$$

$$\sum_{p \in P} x_p \leq B$$

$$x \succcurlyeq 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding x to the original export vector, C is the country that we are interested in, P is the product space, B is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution x * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the first equal part we try to search for the product that can optimize our objective function by

iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where ECI_p is just the ECI of the country after adding some amount of export increment to product p, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make $M_{cp} = 1$ (i.e., the smallest amount of required export increase).

4.3 Economic Panel

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developed countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

4.5 Interpretation

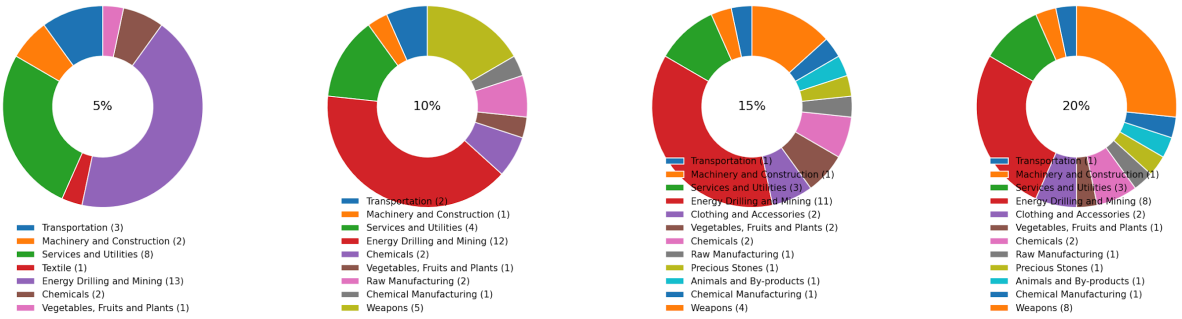
For the purpose of a clear explanation of the line charts, we label all the products in Table 7. Please refer to this table when trying to understand the investment trajectories in the line charts.

Label	Product Names
1	Animal and By-products
2	Vegetables, Fruits and Plants
3	Food and Preparations
4	Energy Drilling and Mining
5	Chemicals
6	Chemical Manufacturing
7	Raw Manufacturing
8	Wood Products
9	Clothing and Accessories
10	Textile
11	Metals
12	Machinery and Construction
13	Transportation
14	Weapons
15	Daily Instruments
16	Precious Stones
17	High-complexity Products
18	Services and Utilities

Table 7. Products Labeling

Figure 4 shows an economic panel for Germany in 2024. Figure 4 shows that ECI gains occur in discrete jumps rather than through a smooth linear increase, suggesting that Germany's complexity improvements depend on reaching certain investment thresholds.

For 2024, a 5 percent increase in export would emphasize Energy Drilling and Mining, followed by Energy Drilling and Mining, Energy Drilling and Mining, and Energy Drilling and Mining for 10, 15, and 20 percent export growth respectively. These strategies go in line with the suggested trajectories based on ECI. The pie charts and Table 8 show that the recommended portfolio is concentrated rather than evenly distributed, with Energy Drilling and Mining repeatedly emphasized across budget scenarios.



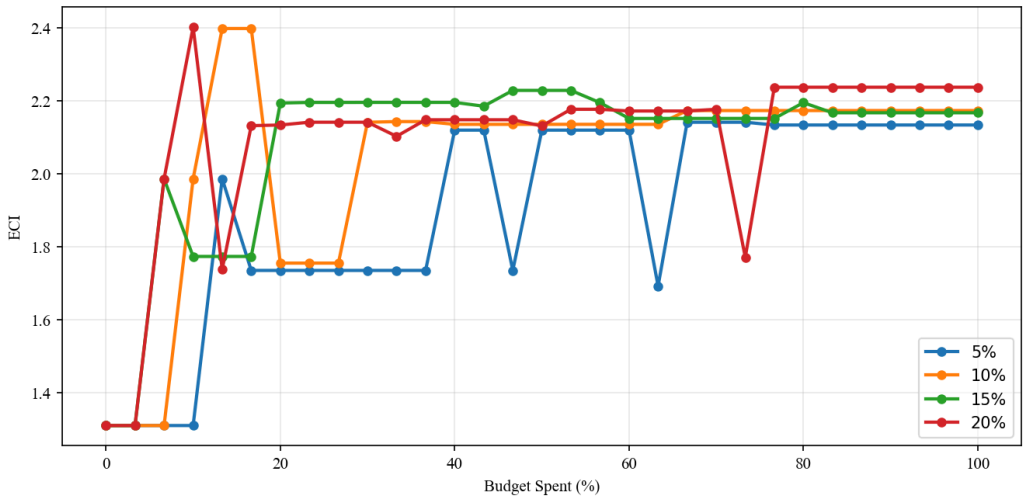


Figure 4. Greedy-Based ECI Improvement Results of Germany with Different Budgets (2024)

3	Chemicals	$6.57B	$13.14B	$19.71B	$26.27B
4	Textile	$3.28B	0	0	0
5	Machinery and Construction	$6.57B	$6.57B	$9.85B	$13.14B
6	Transportation	$9.85B	$13.14B	$9.85B	$13.14B
7	Services and Utilities	$26.27B	$26.27B	$29.56B	$39.41B
8	Chemical Manufacturing	0	$6.57B	$9.85B	$13.14B
9	Raw Manufacturing	0	$13.14B	$9.85B	$13.14B
10	Weapons	0	$32.84B	$39.41B	$105.09B

Table 8. Portfolio of Germany (2024)

4.6 Sensitivity Analysis and Contingency Strategy

In the following section, our broad aim is to understand how sensitive Germany is to changes in export in various industries. While the Economic Complexity Index has been shown to contain valuable information about the economic strength/capacity of a country, little has been said as to the stability of the measure itself. We seek to uncover the main drivers of stability under a country's ECI score and better understand longer-term projections for ECI growth. Additionally, gaining an intuition on the industries that are bolstering an economy and the industries that offer the greatest return per unit of investment is best done through visualization of industry-specific sensitivity. To this end, we have developed two categories of sensitivity analysis.

4.7 ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry’s export level by a certain percentage and recalculate the ECI, all

else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country's level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

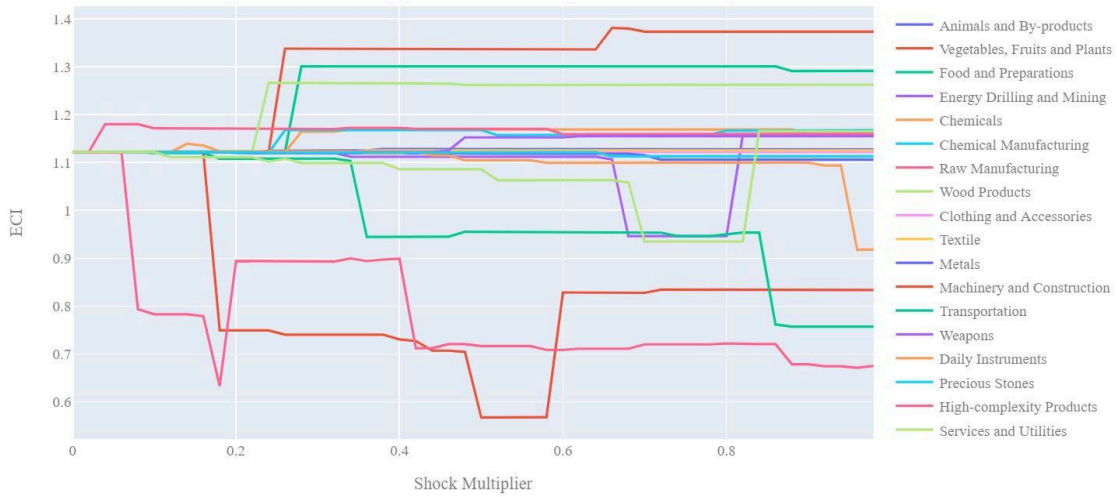


Figure 5. Sample ECI Sensitivity Graph

The graphs are formulated as follows. The first chart, Incremental ECI Sensitivity, shows positive shocking in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

4.8 Sensitivity Analysis Results and Interpretation

We examine Germany's ECI Decremental and Incremental Sensitivity Charts for2024. An increase in Transportation will lead to the highest change in ECI,followed by Machinery and Construction and Metals. Under positive shocks, ECI decreases with additional investment in Transportation, confirming it as the primary lever for complexity growth. For negative shocks, High-complexity Products emerges as the most impactful industry, while reducing investment in Transportation causes ECI to move downward. The rest of the industries do not appear to have the same level of sensitivity.

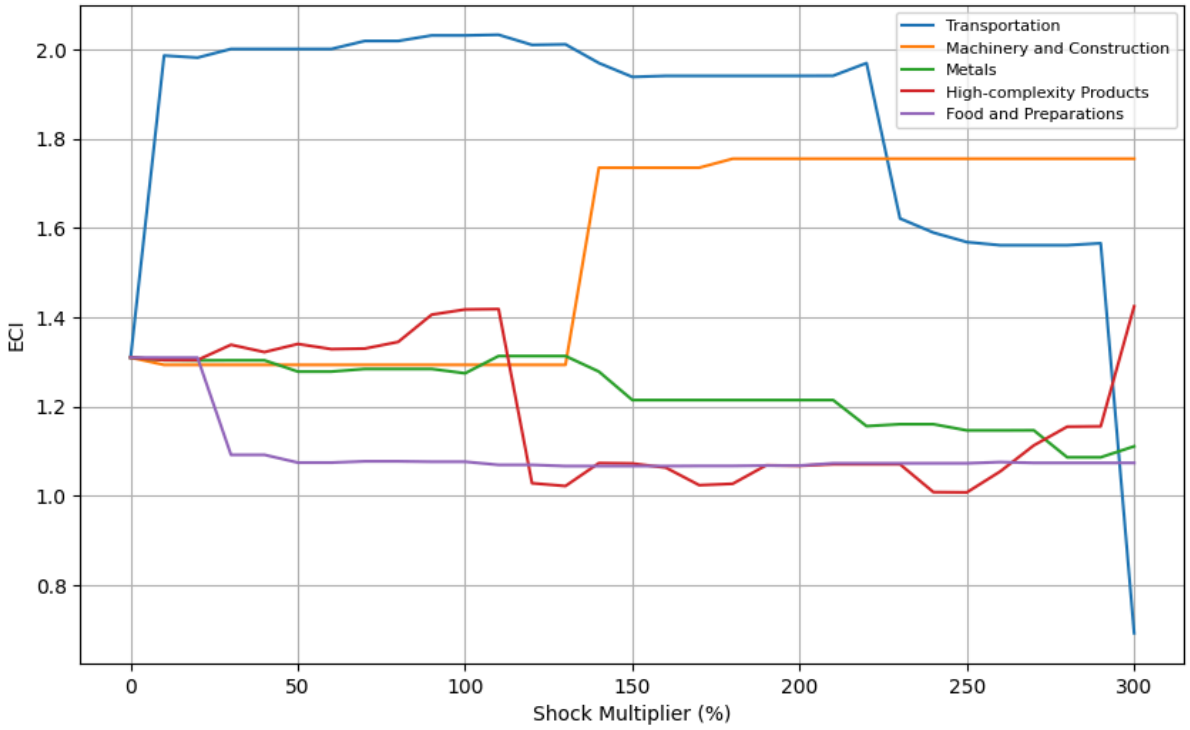


Figure 6a. Incremental ECI Sensitivity for Germany (2024)

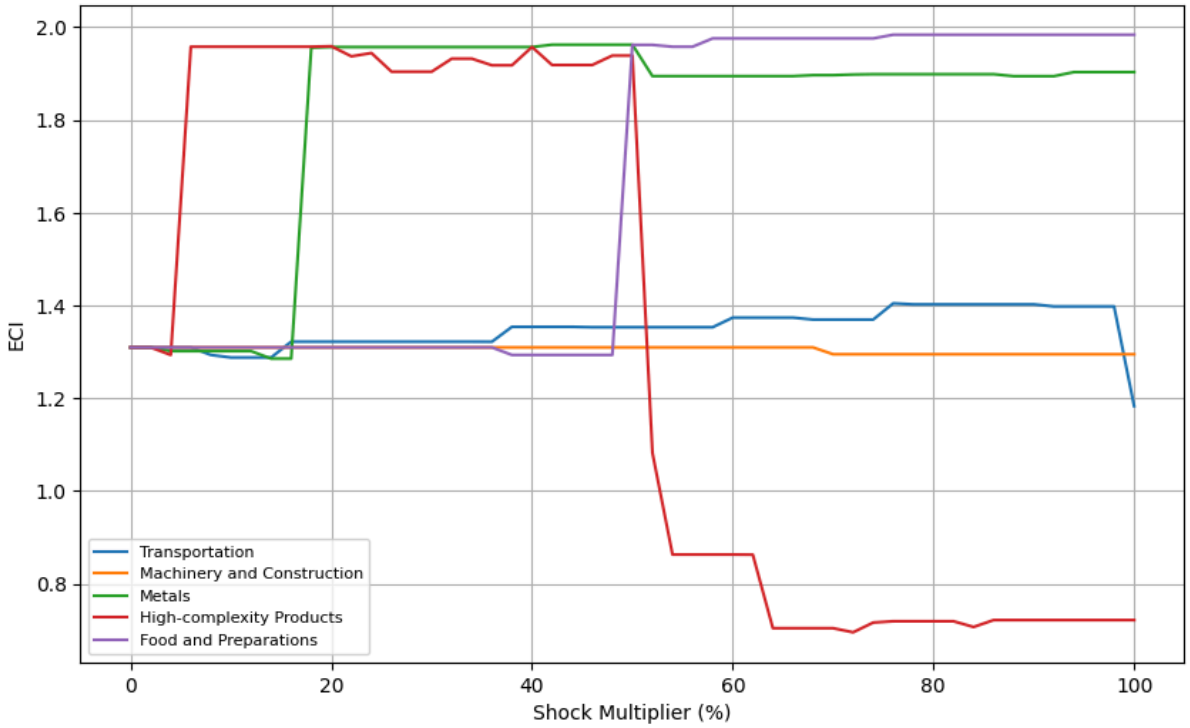


Figure 6b. Decremental ECI Sensitivity for Germany (2024)

5. Macro-level Predictions

5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how

effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing export in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which is defined as the percent increase in GDP (USD), over their SOI. Figure 9 illustrates the regression output, which shows a strong positive correlation between countries' SOI and their five-year GDP growth, achieving an adjusted R^2 of 0.0036%. In other words, the more a country develops, as suggested by calculations on a complexity level, the better economic outcomes it achieves.

Table 14: OLS Regression Results	
Dep. Variable: GDP	**F-Statistic:** 1.4978
R^2: 0.0109	**Prob (F-Stat):** 0.2231
Adj R^2: 0.0036	**No. Observations:** 138

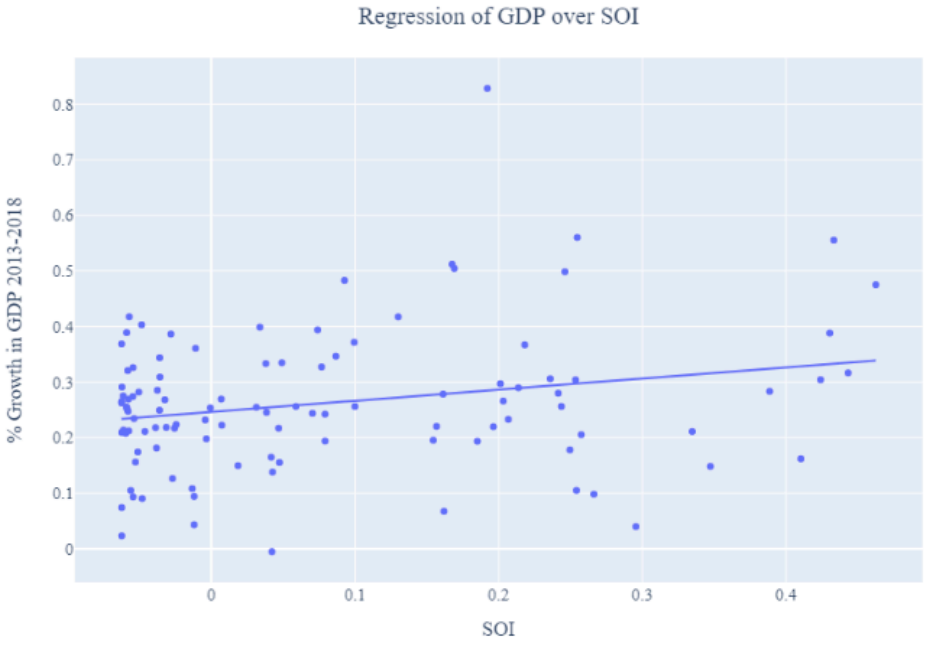


Figure 9. Regression of Five-Year GDP Growth over SOI (2019-2024)

5.2 SOI of Germany and Its Implications

During the examined period, Germany has a high level of structural optimality as compared with all countries for which data is available. As the regression in Figure 10 suggests (the red

curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Germany demonstrates a high level of correlation with the suggested developmental path, having an adjusted R-square of around 0.0036.

Germany ranks in the Q4 quartile among all countries in terms of SOI
and recorded an average annual GDP growth rate of 3.0% from 2019 to 2024, placing it below its developed peers — consistent with the regression result in Figure 9. A useful contrast is the United Kingdom, which recorded a higher average annual GDP growth rate than Germany in the peer comparison and also showed a larger ECI increase over the same period.

Specifically, Germany's medium economic progress from 2019 to 2024 can be at least partially explained by a development strategy that demonstrates a high level of alignment with our suggested developmental path. In particular, Germany has lightly channeled resources toward None — sectors where the complexity opportunity is more limited — while underinvesting in Chemicals, Wood Products, Metals, Machinery and Construction, High-complexity Products, Transportation, Weapons, Daily Instruments, Services and Utilities, which offer greater complexity upside.

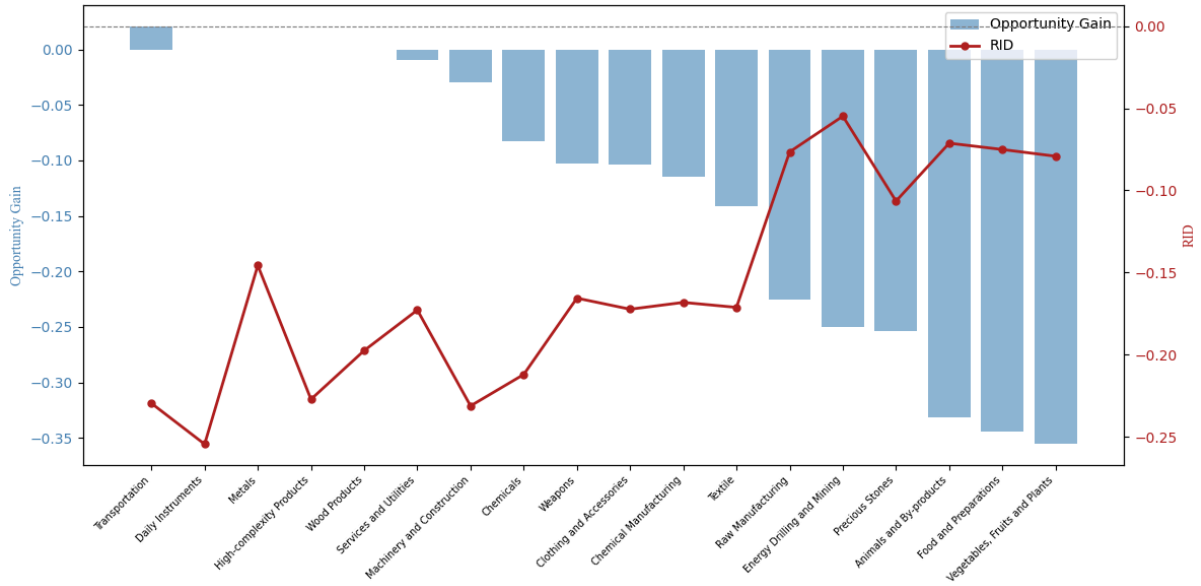


Figure 10. SOI of Germany (2019-2024)

6. Concluding Remarks

In conclusion, Germany has demonstrated a medium level of economic growth in terms of annual GDP increase during the period from 2019 to 2024, concurrent with a significant shift in economic complexity as compared with its peer economies. An average annual GDP growth rate of 3.0% placed Germany below its developed peers, and ECI increased by +4.144 over the same window.

We believe that Germany's trajectory can be partially explained through the analysis of the economic and product complexity structure on the industry group level. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that Germany regressed in developing the majority of its industries during the period, with the count of positively developing industries decreasing from 16 to 0 between 2022 and 2024.

Based on our analysis of opportunity gain and sensitivity across major industry groups, Germany can further its gains by prioritizing Energy Drilling and Mining, Weapons, and Services and Utilities. Sensitivity analysis further identifies Transportation as the sector whose marginal expansion yields the largest positive shift in ECI. We further note that Germany has lightly channeled resources toward None while underinvesting in Chemicals, Wood Products, Metals, Machinery and Construction, High-complexity Products, Transportation, Weapons, Daily Instruments, Services and Utilities, indicating a high level of structural alignment with complexity-guided development, having an adjusted R-square of 0.0036. Since a positive correlation exists between five-year GDP growth and the structural optimality index as we define and calculate, Germany's moderate structural alignment is consistent with its medium growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

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